AGRIVIVE / KHADIN-POD

RAJASTHAN / KHADIN-INSPIRED / SCREENED INFILTRATION CHAMBER

SERIES 21 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



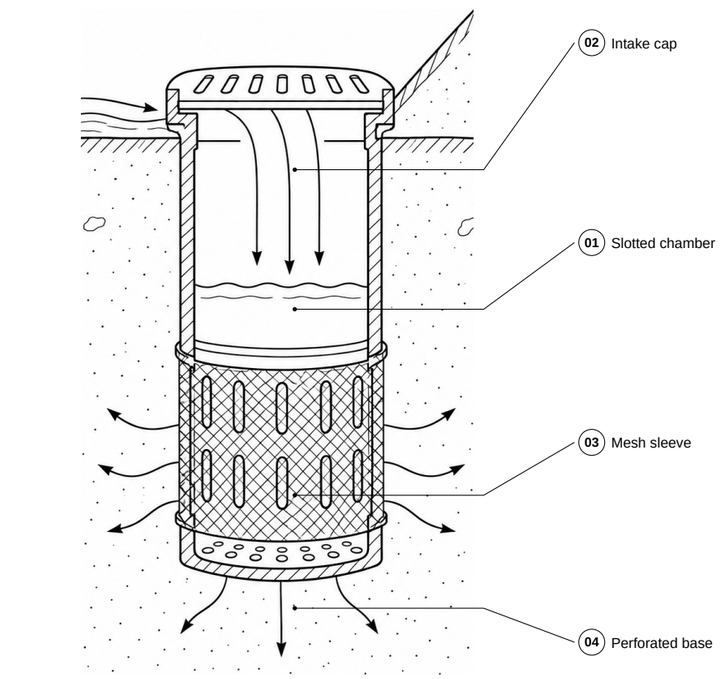
STATE CONTEXT / INDIGENOUS INSPIRATION

Khadins in Rajasthan retain runoff behind field bunds to support soil-moisture use. KHADIN-POD proposes a small screened chamber at a trial bund toe; soil properties determine infiltration and retention.

Practice source: CGWB manual: Khadin runoff farming | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 100 chamber OD x 220 overall H

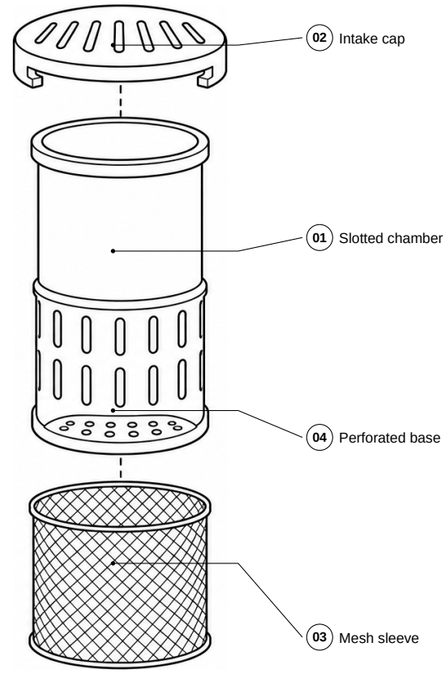


DESIGN / OPERATION NOTES

Prepare a suitable trial pit at the bund toe.
Fit the sleeve around the lower slots and seat the chamber.
Keep intake near runoff grade and remove accumulated sediment through the cap.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
100 chamber OD x 220 overall H

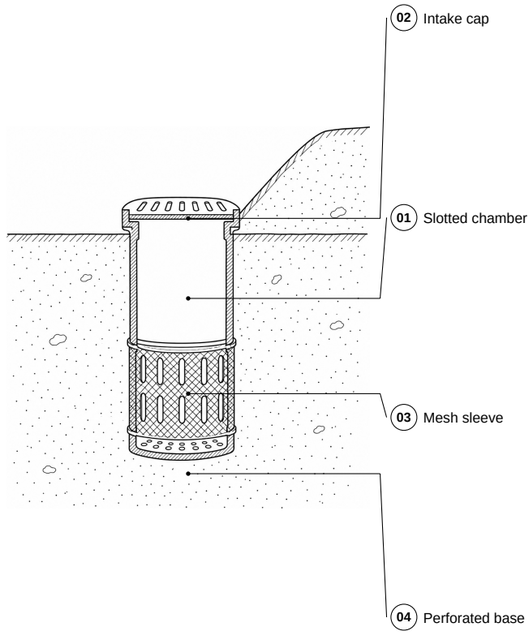


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

SECTION BASIS / mm
90 cavity ID x 180 effective H



DESIGN / OPERATION NOTES

Fit the sleeve around the lower slots and seat the chamber.
Keep intake near runoff grade and remove accumulated sediment through the cap.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Chamber OD / overall H	**100 / 220**
Cavity ID / effective H	**90 / 180**
Screen / slot detail	**TBD**

02 GEOMETRIC CHECK

GROSS CAVITY VOLUME

1,145.1 mL

pi/4 x 90^2 x 180 / 1000
Infiltration and useful retention depend on soil

03 PARTS / QTY

01 Slotted chamber	**1**
02 Intake cap	**1**
03 Mesh sleeve	**1**
04 Perforated base	**Integral**

Chamber/cap: candidate polymer. Soil screen: mesh material and aperture TBD.

04 INTERFACES / CHECKS

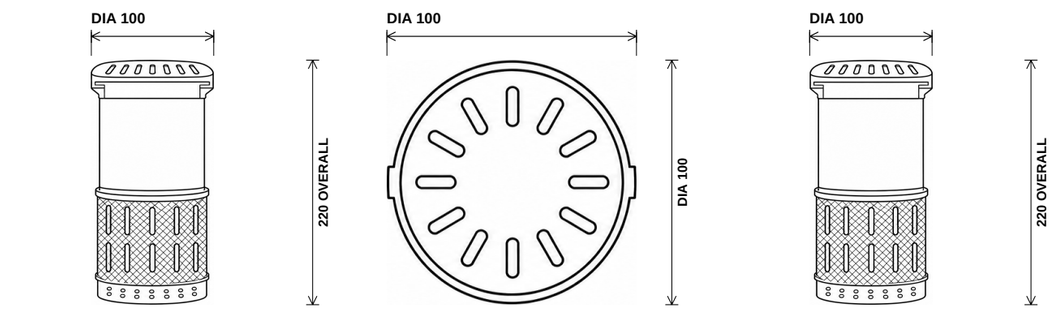
Test infiltration, silting and soil exclusion.
Check cap access and screen retention.
Measure effective void volume after assembly.

05 DATUMS / DRAWING BASIS

A: chamber axis. B: intake-cap rim. Screen covers lower slot band.

PRELIMINARY / VALIDATION PENDING

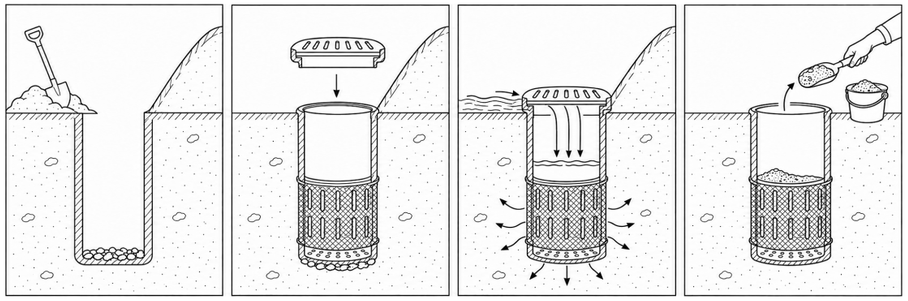
03 ORTHOGRAPHIC STUDIES / mm



FRONT **TOP** **SIDE**

Chamber DIA 100 OD x 220 H. Gross clear cavity DIA 90 x 180 effective H; usable volume lower.

05 INSTALLATION / OPERATING SEQUENCE



1 PREPARE PIT **2 FIT SLEEVE** **3 SEAT / CAP** **4 CLEAR SEDIMENT**

Concept sequence; validate fit, access and operating conditions in prototypes.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-21 / P01	KHADIN-POD	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	21 / 28